Summarizing Distributions with Center, Spread and Shape Answer Key

Precalculus / Statistics

Quick Answers

1. 40 minutes
2. 37.5 minutes
3. 65 minutes
4. (a) the week's total = 95, (b) the busiest day's count = 27
5. (a) Q1 = 30 minutes, (b) Q3 = 45 minutes, (c) the IQR = 15 minutes
6. (a) the upper fence = 67.5 minutes, —
7. (a) the mean = 38 minutes, (b) the median = 38 minutes, —
8. 8 minutes
9. $\frac{1}{5}$
10. (a) the new mean = 36 minutes, (b) the new median = 37.5 minutes, —

What is verified

15 of 18 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 6, 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus / Statistics
HSS-ID.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

2. *If they got 35: took the middle entry of a ten-value list instead of averaging the two middle entries*

Class A (minutes): $20, 25, 30, 30, 35, 40, 40, 45, 50, 85$
Class B (minutes): $26, 30, 34, 38, 42, 46, 50$

1. Find the mean daily reading time for Class A.

Solution: The ten values total 400, so the mean is $\frac{400}{10}$. $\boxed{\text{40 minutes}}$

2. Find the median daily reading time for Class A.

Solution: Ten values, so the median is the average of the 5th and 6th: $\dfrac{35+40}{2}$.

$\boxed{37.5 \text{ minutes}}$

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3. Find the range of the Class A times.

Solution: Largest minus smallest: $85 - 20$. $\boxed{65 \text{ minutes}}$

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4. Library borrowings Mon–Fri: $12, 19, 15, 22, 27$. (a) week total. (b) busiest day's count.

Solution (a): $12 + 19 + 15 + 22 + 27$. $\boxed{95}$

Solution (b): The largest count is Friday's. $\boxed{27}$

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5. Class A quartiles. (a) Q_1. (b) Q_3. (c) IQR.

Solution (a): Split at the median: the lower half is $20, 25, 30, 30, 35$, whose middle value is the third. $\boxed{Q_1 = 30 \text{ minutes}}$

Solution (b): The upper half is $40, 40, 45, 50, 85$, whose middle value is the third. $\boxed{Q_3 = 45 \text{ minutes}}$

Solution (c): $45 - 30$. $\boxed{\text{IQR} = 15 \text{ minutes}}$

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6. Outliers for Class A. (a) upper fence $Q_3 + 1.5 \times \text{IQR}$. (b) name the outlier and describe the shape.

Solution (a): $45 + 1.5(15) = 45 + 22.5$. $\boxed{67.5 \text{ minutes}}$

Solution (b) — instructor-judged. Only 85 exceeds 67.5, so 85 is the single high outlier. (The lower fence, $30 - 22.5 = 7.5$, is below every value, so there is no low outlier.) One long right tail pulls the mean of 40 above the median of 37.5: the distribution is right-skewed, with center near 37.5–40 and a middle half only 15 wide. Full credit requires naming 85 as the outlier and calling the shape right-skewed with the mean-above-median reason.

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7. Class B. (a) mean. (b) median. (c) what their agreement says about shape.

Solution (a): The seven values total 266, and $\dfrac{266}{7} = 38$. $\boxed{38 \text{ minutes}}$

Solution (b): Seven values, so the median is the 4th. $\boxed{38 \text{ minutes}}$

Solution (c) — instructor-judged. Mean and median are both 38. When the two agree, no tail is pulling the balance point away from the middle position, so the distribution is roughly symmetric — and indeed the values are evenly spaced about 38. Class A is the contrast: there the mean (40) sits above the median (37.5) because of the outlier at 85. Full credit requires the symmetry conclusion and the comparison with Class A.

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8. Find the population standard deviation of the Class B times.

Solution: Deviations from the mean of 38 are $-12, -8, -4, 0, 4, 8, 12$; their squares total 448; the variance is $\frac{448}{7} = 64$, and $\sqrt{64} = 8$. $\boxed{\text{8 minutes}}$

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9. One of the ten Class A students is chosen at random. Find P(read more than 45 minutes).

Solution: Two of the ten values exceed 45 (namely 50 and 85), so $P = \frac{2}{10}$. $\boxed{P = \frac{1}{5}}$

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10. **Synthesis.** The 85 is corrected to 45, giving $20, 25, 30, 30, 35, 40, 40, 45, 45, 50$. (a) new mean. (b) new median. (c) why one moved and the other did not.

Solution (a): The corrected values total 360, so the mean is $\frac{360}{10}$. $\boxed{\text{36 minutes}}$

Solution (b): The 5th and 6th values are still 35 and 40, so the median is $\frac{35+40}{2}$. $\boxed{\text{37.5 minutes}}$

Solution (c) — instructor-judged. The mean fell from 40 to 36; the median did not move at all. The mean is built from the total, so lowering one value by 40 lowers the total by 40 and the mean by $40/10 = 4$. The median depends only on which values sit in the middle two positions, and correcting the largest value left that pair untouched — the median is *resistant* to extreme values. This is why a skewed distribution is usually summarized by the median and IQR rather than the mean and standard deviation. Full credit requires the observed movement of each measure and the resistance argument.